

Wenqi (Wendy) Xian

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[google scholar profile](#), [LinkedIn profile](#)

EDUCATION

Cornell Tech, New York, NY

Ph.D. in Computer Science

2018 – 2023

Advisor: Prof. Noah Snavely

Georgia Institute of Technology, Atlanta, GA

B.S. in Computer Science

2014 – 2018

RESEARCH INTERESTS

My research focuses on AI for creative applications, from lights, to camera, and action. I'm interested in controllable, multi-modal video diffusion models that anyone can actually direct, not just prompt.

EXPERIENCES

Luma AI

Oct. 2025 – Now

Research Scientist

Redwood City, CA

- Core contributor to Luma's flagship Ray 3.x video models, specialize in post-training and evaluating multimodal models on large-scale video data.
- Built video editing models that achieved state-of-the-art subject consistency and precise performance transfer, shipping production-ready features and agentic workflow to large commercial studios.

Netflix Eyeline Studios

August. 2023 – Oct. 2025

Research Scientist

Los Angeles, CA

- Advanced photorealistic reconstruction with 3D/4D Gaussian Splatting, significantly improving efficiency and reducing costs compared to traditional photogrammetry pipeline.
- Enhanced controllable video generation through implementing features of precise object and camera motion, relighting, personalization, and keyframing.
- Developed state-of-the-art models of video depth estimation and video super-resolution, for real-time composition in virtual film production.

Meta Reality Labs

May. 2022 – Aug. 2022

Research Intern

Sausalito, CA

Designed a neural camera model for rendering and optimizing scene representations end-to-end from raw photos and unknown camera parameters.

Meta Reality Labs

Aug. 2021 – Nov. 2021

Research Intern

Remote

Proposed a novel method to drive 2D image generation from 3D view point changes, enabling perpetual view generation from a single input image.

Meta Reality Labs

May. 2020 – Aug. 2020

Research Intern

Seattle, WA

Extended NeRF to dynamic scenes through video depth estimation for 3D video creation.

Adobe Research

May. 2019 – Aug. 2019

Research Intern

Seattle, WA

Proposed to improve video object removal by leveraging 3D reconstruction and geometry.

Adobe Research

Research Intern

Automated single image camera calibration for 3D object insertion in Adobe Dimension.

May. 2018 – Aug. 2018

San Francisco, CA

UC Berkeley

Research Intern

Built an annotation tool and collected a large-scale driving dataset of 100K videos to facilitate object detection and tracking in autonomous driving research.

May. 2017 – Aug. 2017

Berkeley, CA

PUBLICATIONS

Detail Enhanced Gaussian Splatting for Large-Scale Volumetric Capture

Philip J.*, Ma L.*, Clausen P.*, **Xian W.***, Taşel A. L., He M., Yu X., George D. M., Yu N., Pilarski O., Debevec P. *co-first author
SIGGRAPH Asia 2025

Virtually Being: Customizing Camera-Controllable Video Diffusion Models with Multi-View Performance Captures

Xu Y., **Xian W.**, Ma L., Philip J., Taşel A., Zhao Y., Burgert R., He M., Hermann O., Pilarski O., Garg R., Debevec P., Yu N.
SIGGRAPH Asia 2025

FlashDepth: Real-time Streaming Video Depth Estimation at 2K Resolution

Chou G., **Xian W.**, Yang G., Abdelfattah M., Hariharan B., Snavely N., Deng Y., Yu N., Debevec P.
*ICCV 2025 *Spotlight**

Go-with-the-Flow: Motion-Control Video Diffusion Models Using Real-Time Warped Noise

Burgert R., Xu Y., **Xian W.**, Pilarski O., Clausen P., He M., Ma L., Deng Y., Li L., Mousavi M., Ryoo M., Debevec P., Yu N. *CVPR 2025 *Oral**

Lux Post Facto: Learning Portrait Performance Relighting with Conditional Video Diffusion and a Hybrid Dataset

Mei Y., He M., Ma L., Philip J., **Xian W.**, George D., Yu X., Debic G., A. Tasel, Yu N., V Patel., Debevec P. *CVPR 2025*

Self-Calibrating Gaussian Splatting for Large Field of View Reconstruction

Deng Y., **Xian W.**, Yang G., Guibas L., Wetzstein G., Marschner S., Debevec P. *ICCV 2025 *Spotlight**

DiffRelight: Diffusion-Based Facial Performance Relighting

He M., Clausen P., Taşel A., Ma L., Pilarski O., **Xian W.**, Rikker L., Yu X., Burgert R., Yu N., Debevec P. *SIGGRAPH Asia 2024*

Neural Lens Modeling

Xian, W., Bozic A., Snavely, N., Lassner, C. *CVPR 2023*

FactorMatte: Redefining Video Matting for Re-Composition Tasks

Gu Z., **Xian, W.**, Snavely, N., Davis, A. *SIGGRAPH 2023*

Space-time Neural Irradiance Fields for Free-Viewpoint Video.

Xian, W., Huang J., Kopf, J., Kim, C. *CVPR 2021*

Stay Positive: Non-Negative Image Synthesis for Augmented Reality.

Luo, K., Yang, G., **Xian, W.**, Harald H., Hariharan B., Belongie S. *CVPR 2021 Oral*

Crowdsampling the Plenoptic Function.

Li, Z., **Xian, W.**, Davis, A., Snavely, N. *ECCV 2020 Oral*

BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning.

Yu F., Chen H., Wang X., **Xian, W.**, Chen Y., Liu F., Madhavan V., Darrell T. *CVPR 2020 Oral*

UprightNet: Geometry-Aware Camera Orientation Estimation from Single Images.

Xian, W.*, Li, Z.*, Fisher, M., Eisenmann, J., Schechtman, E., Snavely, N. *co-first *ICCV 2019*

Texturegan: Controlling Deep Image Synthesis with Texture Patches.

Xian, W.*, Sangkloy P*, Agrawal, V., Raj, A., Lu J., Fang C., Yu F., Hays J. *co-first author
CVPR 2018 Spotlight Oral

AWARDS & FELLOWSHIP

Microsoft Ada Lovelace Research Fellowship 2020	2020
World Second Place Winner of Microsoft Imagine Cup 2017	2017
President's Undergraduate Research Award	2017

SERVICE

Paper reviewer of ICCV, CVPR, ECCV, SIGGRAPH, NeurIPS etc.

Teaching assistant: Deep Learning (Spring 2019), Virtual & Augmented Reality (Fall 2019),
Computer Vision (Spring 2020, Spring 2022)